

Junghyeok Lee

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RESEARCH INTERESTS

- Memory-Centric and Processing-in-Memory Architectures for AI Systems
- Design-Space Exploration Frameworks for Heterogeneous LLM Serving
- Hardware-Aware LLM Compression and Inference Acceleration for On-Device AI

EXPERIENCES

- Researcher, SoC Platform Research Center, KETI (Korea Electronics Technology Institute), Korea (Jan. 2026 ~ Aug. 2026)
- Researcher Student at Intelligent Digital System design Lab, Seoul National University of Science and Technology, Korea (Jul. 2023 ~ Sep. 2023)

EDUCATION

Sep. 2026 ~ Present	Seoul National University of Science and Technology Department of Electrical and Information Engineering <i>Ph.D. student / Advisor: Hyun Kim</i>	Seoul, Korea
Sep. 2023 ~ Aug. 2025	Seoul National University of Science and Technology Department of Electrical and Information Engineering <i>Master of Science / Advisor: Hyun Kim</i>	Seoul, Korea
Mar. 2016 ~ Feb. 2023	Hankuk University of Foreign Studies Department of Electronic Physics <i>Bachelor of Science</i>	Yongin, Korea

CONFERENCES

- **Junghyeok Lee***, JiHoon Jang*, and Hyun Kim, “PriME: PIM-Aware Efficient Compression for Memory-Bound Embedding Layers in sLLMs,” *2025 IEEE 43rd International Conference on Computer Design (ICCD)*. IEEE, 2025. (IF: 1, NRF BK21+ Acceptance Rate: 25.5%)
- **Junghyeok Lee**, JiHoon Jang, and Hyun Kim, “XNC: XOR and NOT-Based Lossless Compression for Optimizing Unquantized Embedding Layers in Large Language Models,” *2025 IEEE International Symposium on Circuits and Systems (ISCAS)*. IEEE, 2025.

JOURNALS

- Inseong Hwang*, **Junghyeok Lee***, Huibeom Kang, Gilhyeon Lee, and Hyun Kim, “Survey of CPU and Memory Simulators in Computer Architecture: A Comprehensive Analysis Including Compiler Integration and Emerging Technology Applications,” *Simulation Modelling Practice and Theory* (2025): 103032.

CONFERENCES - DOMESTIC

- **이정혁**, 김병수, 전석훈, “AiM-CoSim: GDDR6-AiM Timing-Functional Co-Simulation 프레임워크”, *대한전자공학회 하계학술대회 (2026)*
- 이채린, **이정혁**, 전석훈, “LLM 추론에서 이기종 연산기의 수치 오차 허용 한도 탐색을 위한 에러 모델”, *대한전자공학회 하계학술대회 (2026)*
- **이정혁**, 장지훈, 김현, “VLM의 시각-언어 토큰 modality 차이에 따른 양자화 정확도 저하 완화를 위한 무손실 압축 기법,” *대한전자공학회 학술대회 (2025): 2852-2854*.
- **이정혁**, 장지훈, 김현, “메모리 접근 효율을 고려한 트랜스포머 기반 모델들의 실제 데이터 압축률 분석,” *대한전자공학회 학술대회 (2024): 687-689*.

PATENTS

1. 김현, **이정혁**, 장지훈, “임베딩 레이어의 압축을 통한 메모리를 기반으로 저전력 처리를 위한 전자 장치 및 그 구동 방법”, 한국, 서울과학기술대학교 산학협력단, 10-2025-0065237

AWARDS AND HONORS

- 이길현, 윤석규, 김영찬, **이정혁**, 장지훈, "2023-2024 Student Design Competition Winner in Seoul Chapter" 수상, IEEE Circuits and Systems Society, 2024.01.31

SKILLS AND TECHNIQUES

- Verilog
- Python
- C / C++

SCHOLARSHIP/SUPPORTS

- 인공지능 반도체 융합 전문 인력 육성 사업(Multidisciplinary Research Training and Development Enterprise for AI and Semiconductor Technology), 2020.04~2025.12, 한국연구재단 지원

PROJECTS

- 2T DRAM 기반 저전력, 고성능 PIM 기술을 위한 회로 및 아키텍처 개발(Development of circuits and architecture for 2T DRAM-based low-power and high-performance PIM cells), 2022.04~2024.12, 한국연구재단 지원
- 초거대 언어모델을 위한 양자화 기술 및 스케줄링 기법을 포함하는 Transformer 가속 솔루션 개발 (Development of Transformer Acceleration Solution Including Quantization and Scheduling Techniques for LLM), 2024.04~2026.12, 산업통상자원부 지원
- 시뮬레이션 기반 고속/고정확도 데이터센터 워크로드/시스템 분석 플랫폼 개발, 2026.01~2026.09, 과기정통부 지원